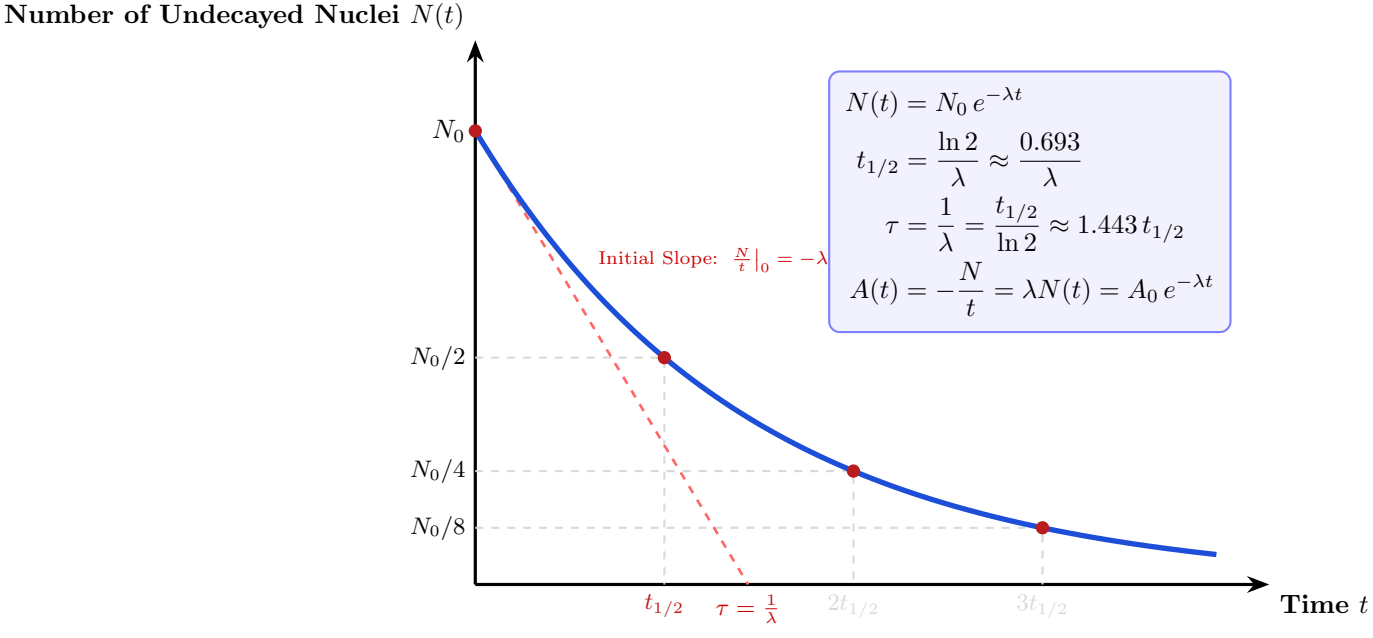

Number of Undecayed Nuclei $N(t)$
N_0
$N_0/2$
$N_0/4$
$N_0/8$
Initial Slope: $\frac{N}{t}\big|_0 = -\lambda$
$N(t) = N_0\, e^{-\lambda t}$
$t_{1/2} = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda}$
$\tau = \frac{1}{\lambda} = \frac{t_{1/2}}{\ln 2} \approx 1.443\, t_{1/2}$
$A(t) = -\frac{N}{t} = \lambda N(t) = A_0\, e^{-\lambda t}$
$t_{1/2}$
$\tau = \frac{1}{\lambda}$
$2t_{1/2}$
$3t_{1/2}$
Time t